

Central Sapporo (Clock Tower & Sosei River)

Then-and-now aerial photographs, how the ground was made, and the disasters people chose to remember — compiled automatically from public Japanese government data.



札幌都心（時計台・創成川）

Study area	2.2 km × 1.4 km around 43.0615° N, 141.3540° E
Oldest aerial photograph	1945–1950
Aerial series in this brief	1945–1950, 1961–1969, 1974–1978, 1984–1986, 2020
Main natural landform	Alluvial fan (93%)
Former river channels	3.1%
Filled, reclaimed or drained land	none mapped
Former sea, river or pond	none mapped
Meiji-era paddies, marsh and flats	none mapped
Disaster memorials within 3 km	0

What this brief is. A reading aid for comparing the same ground across decades. It shows what the photographs and GSI's land surveys record; it does not describe current conditions, access or opening hours.

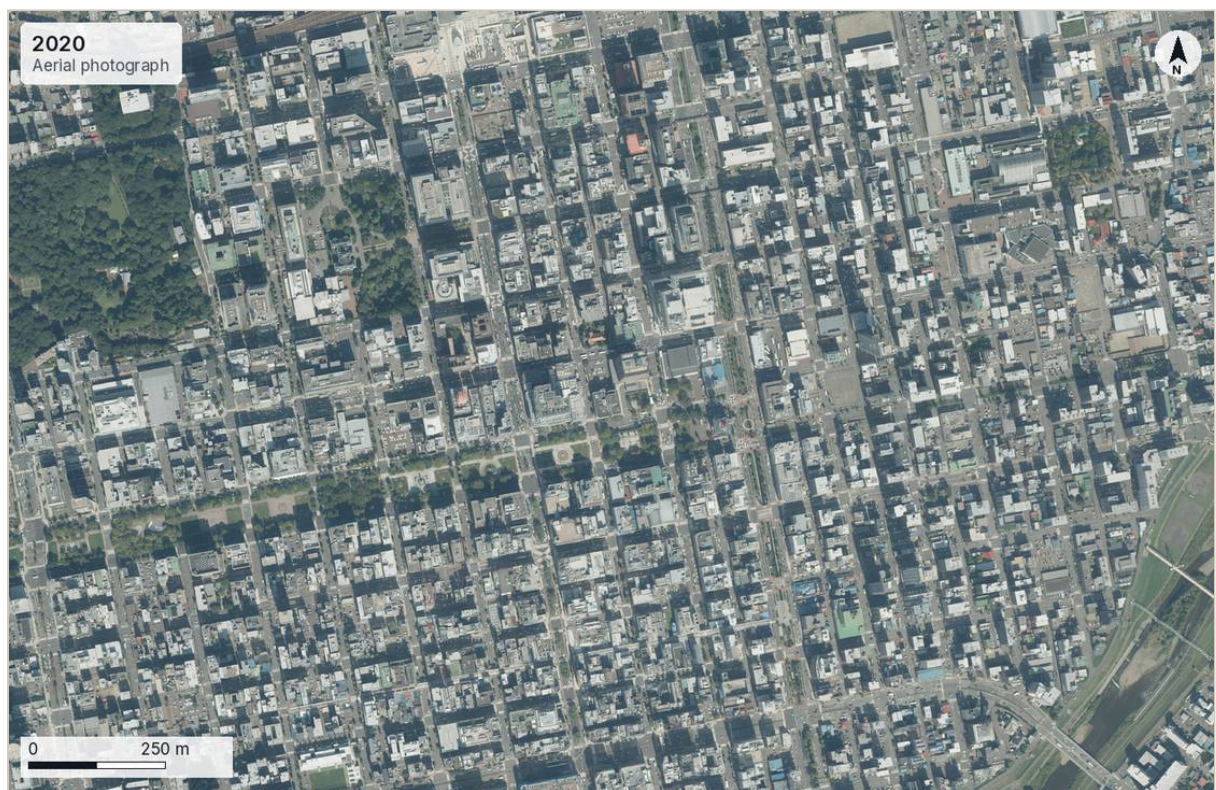
What it is not. Not a hazard assessment or a property survey. Landform classes describe typical tendencies of the ground; for present-day risk, read the official hazard map of the municipality.

Edition 2026-09-27. Aerial photographs, landform and Meiji-era lowland data: Geospatial Information Authority of Japan (GSI), processed by Japan Time Atlas. Memorial data: GSI Natural Disaster Memorials, summarised and translated by Japan Time Atlas.

The same ground, 1945–1950 and 2020



1945–1950 aerial photograph (US military photography) — GSI Tiles *ort_USA10*, cropped and annotated by Japan Time Atlas. North is up; the frame is identical in both images.



2020 annual orthophoto — GSI Tiles *nendophoto2020*, cropped and annotated by Japan Time Atlas. Capture dates within a mosaic can differ from place to place.

How to compare. Anchor on features that rarely move — river banks, rail lines, shrine and temple grounds, main roads — then look for what changed around them. Differences in tone come from film, camera and season, not from the land itself.

The decades in between

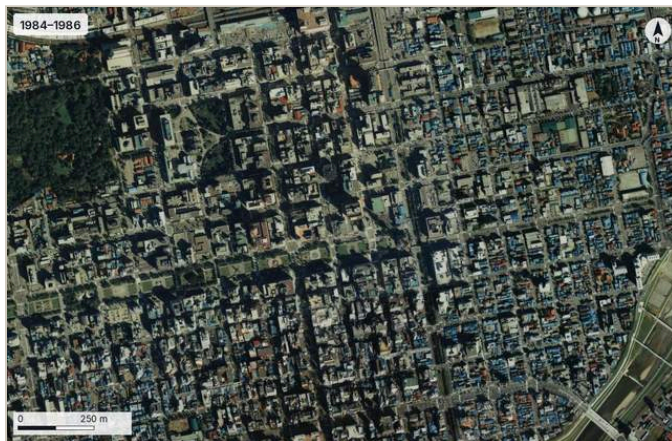
Every further GSI series that covers the same frame, oldest first. Each series is a compilation: photographs within one series were taken on different dates, which is why seams and changes of tone can appear inside a frame.



1961-1969 · GSI Tiles *ort_old10*, processed



1974-1978 · GSI Tiles *gazo1*, processed



1984-1986 · GSI Tiles *gazo3*, processed

Rivers, shorelines and terraces



Natural landforms as classified by GSI, over a recent aerial photograph in gray. Dark blue outlines mark former river channels.

 Alluvial fan 93%	 Former river channel 3.1%
 Water 1.4%	 Floodplain and coastal plain 0.6%

What the main landforms mean

Alluvial fan (93%). Gently sloping ground spreading fan-like from a valley mouth, built from sediment dropped by floods. Floodwater from the mountains, and debris flows near the valley mouth. Relatively firm ground; a slight liquefaction tendency on the lower fan.

Former river channel (3.1%). A former course of a river, slightly lower than its surroundings, cut off when the river moved and later filled with sand and mud. Floodwater stays longer and drains poorly; soft ground amplifies shaking; where filled or raised, the liquefaction tendency is very strong.

Descriptions translated and condensed from GSI's landform-classification legend. They describe tendencies of each landform type, not the condition of any individual plot.

Fill, reclamation and the fields beneath the town



Hatching: man-made ground as classified by GSI. Teal outlines: former sea, river or pond. Dark blue outlines: former river channels.

Man-made ground

No man-made ground mapped in this frame.

Meiji-era lowland

GSI has not traced Meiji-era lowland for this frame.

Memorials to natural disasters nearby



Numbered memorials; the black rectangle is the study area of this brief. Frame 60 km across.

Fewer than three memorials stand within 3 km of the study area, so the nearest ones within 30 km are listed.

Memorial	Disaster	Erected	Distance
1 Monument of gratitude for the Yubari River flood-control works 治水感謝の碑	Floods of 7 Sep 1898 and Aug–Sep 1932, and others · flood	2004	26.3 km

What the memorials record

1. **Monument of gratitude for the Yubari River flood-control works.** The Yubari River caused severe disasters, including the floods of 1898 and August 1932. Flood-control works begun in 1920 and completed in 1936 turned the area into a town where people can enjoy the water in peace. The monument thanks those who carried out the works; beside it stands another honouring their achievement.

Source: GSI Natural Disaster Memorials (自然災害伝承碑). English names and notes are Japan Time Atlas translations and summaries of GSI's Japanese descriptions; the memorials' own inscriptions may say more. "Erected" is the year GSI records for the monument.

Where every element comes from

- **Aerial photographs** (pp. 2–3) — Geospatial Information Authority of Japan (GSI), GSI Tiles: series *ort_USA10* (1945–1950, US military photography), *ort_old10* (1961–1969), *gazo1* (1974–1978), *gazo3* (1984–1986) and *nendophoto2020*. Cropped, resized and annotated by Japan Time Atlas. List of GSI Tiles: <https://maps.gsi.go.jp/development/ichiran.html>
- **Landform classification** (pp. 4–5) — GSI landform-classification vector tiles (natural and artificial landforms), compiled by GSI from its Land Condition Maps, Flood Control Terrain Classification Maps and vulnerable-landform surveys. Rendered, measured and translated into English by Japan Time Atlas. https://github.com/gsi-cyberjapan/experimental_landformclassification
- **Meiji-era lowland** (p. 5) — GSI “明治期の低湿地” tiles and legend (https://maps.gsi.go.jp/legend/lw_legend.pdf), measured and translated by Japan Time Atlas.
- **Natural disaster memorials** (p. 6) — GSI Natural Disaster Memorials (自然災害伝承碑), registered by municipalities, <https://www.gsi.go.jp/bousaichiri/denshouhi.html>; processed by Japan Time Atlas (selection by distance, English translation and summary). Memorial photographs are not reproduced.
- GSI content is used under the GSI Content Terms of Use, which follow the Public Data License v1.0 (PDL1.0, compatible with CC BY 4.0). Data retrieved 2026-09-27.
- **Outline of Japan** — Data of Japan / JapanPrefGeoJson (public domain), simplified.

Method

The study area is a fixed rectangle of 2.2 × 1.4 km. Every GSI photo series that covers at least 85% of it is included; the newest annual orthophoto covering at least 90% of it is used as “now”, with any gap filled from GSI's latest seamless mosaic. Landform shares are areas of the rectangle measured on GSI's polygons at zoom level 14, each class merged before measuring. Meiji-era shares count map pixels by legend color. Memorials are selected by straight-line distance from the center of the study area.

Limits

Historical series are mosaics of photographs taken on different dates; GSI publishes the flight date of each photograph. Landform data record conditions at the time of survey. Nothing in this brief shows present-day access, safety or opening hours.

Area Dossier. The same district with a 4–8 page English history, written from Japanese library sources and cited page by page.

Deep Research. A single hamlet or address, researched in libraries and archives.

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